Data Journalism

Session 4: Collecting data: web, APIs, open sources

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Today

1. Data and newsworthiness
2. Data sources for journalism
3. Resources

Data and newsworthiness

What makes data newsworthy?

Dimensions of data newsworthiness

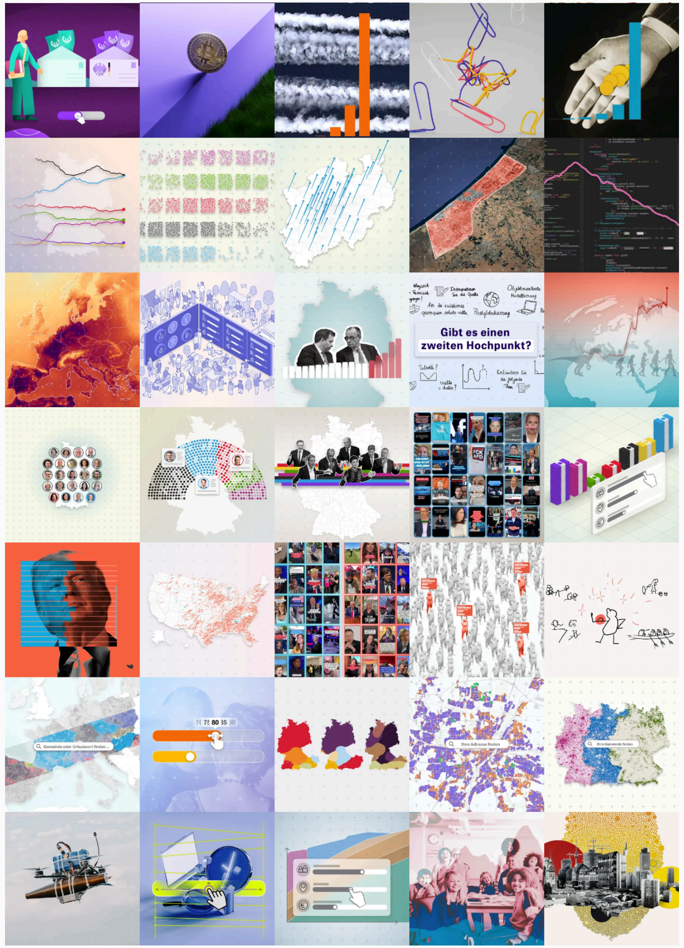
Surprise — Does the data contradict what officials claim, what the public believes, or what previous research found?

Scale — Does it quantify something that was previously only anecdotal? Can it show that a problem is bigger (or smaller) than assumed?

Disparity — Does it reveal unequal treatment, access, or outcomes across groups?

Trend — Does it show something getting worse or better over time in ways that matter?

Comparability — Does it help the reader understand something about themselves?



`Source` ZEIT, Daten und Visualisierung

Two routes into a data story

Route 1: Data first

You stumble upon a dataset that is genuinely curious — unexpected patterns, suspiciously clean numbers, a leaked dataset of AI-generated content that shows systematic bias against certain topics.

The dataset is the tip. Follow it.

Route 2: Story first

Public discourse is already moving — migration, inflation, housing, AI surveillance. Ask: which data could tell me something useful about this that nobody has looked at yet?

The question is the tip. Find data that can answer it.

The journalist's instinct

Not every dataset is a story. The question is whether the data reveals something surprising, consequential, or hidden — ideally all three.

A useful exercise

Take today's front page. For each story: what data exists? What data is *missing*? Who controls it?

Data sources for journalism

1. Open government data

What it is

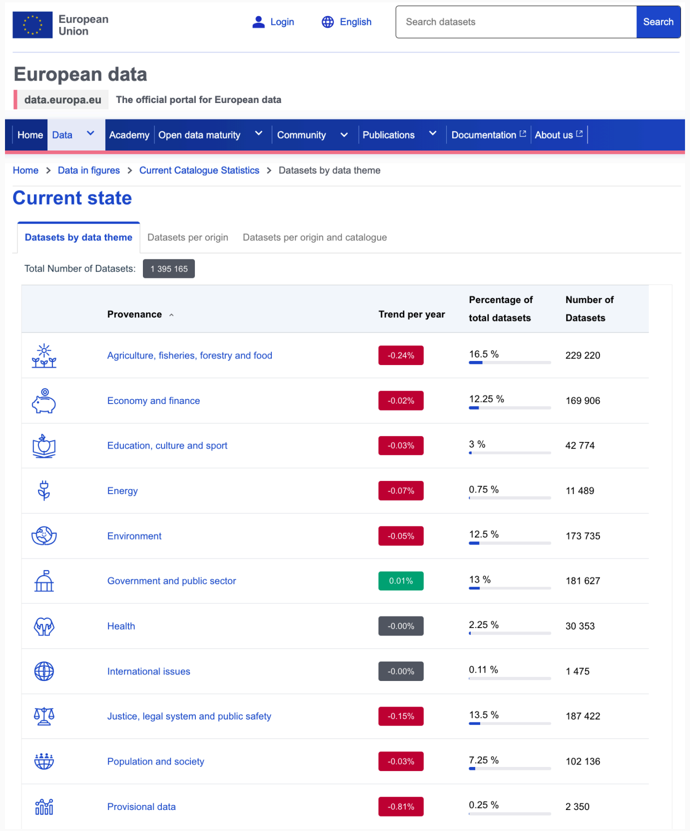
Data published by governments as a matter of policy or legal obligation — usually without restrictions on use.

Some key portals

- data.europa.eu — EU open data portal
- data.gov — US federal data
- govdata.de — Germany
- TED, USASpending.gov, Eurostat, OECD.Stat, World Bank Open Data

What to watch for

- Data quality varies enormously
- "Open" doesn't mean easy: PDFs, inconsistent formats, missing codebooks. But this can also increase the value of your work
- Publication lag: some datasets are months behind reality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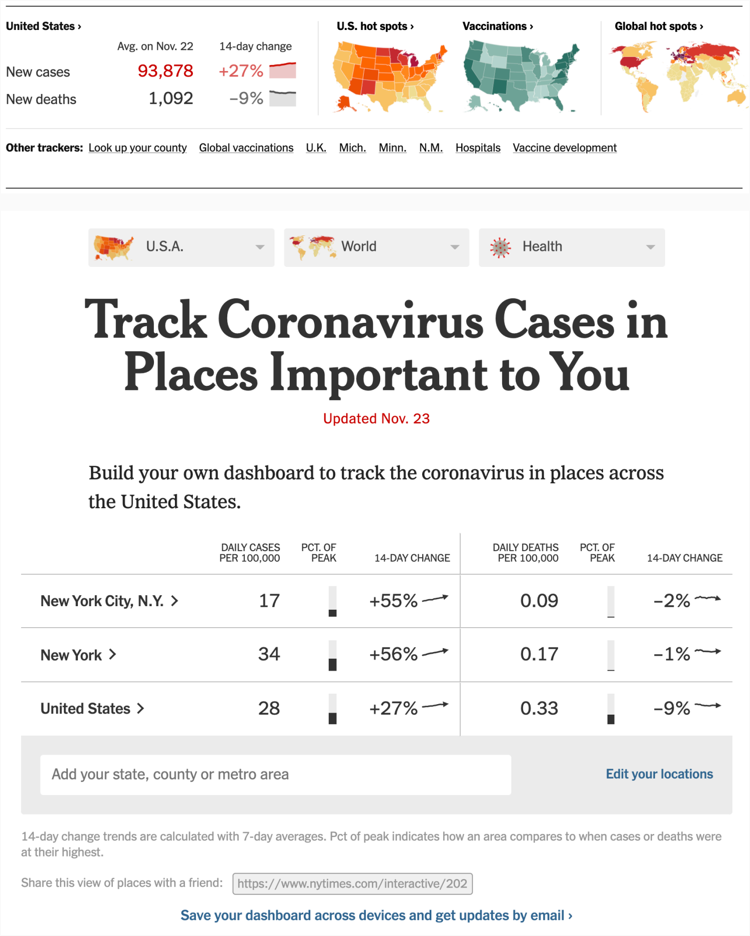
Current state

Datasets by data theme | Datasets per origin | Datasets per origin and catalogue

Total Number of Datasets: 1 395 165

Provenance	Trend per year	Percentage of total datasets	Number of Datasets
Agriculture, fisheries, forestry and food	-0.24%	16.5 %	229 220
Economy and finance	-0.02%	12.25 %	169 906
Education, culture and sport	-0.03%	3 %	42 774
Energy	-0.07%	0.75 %	11 489
Environment	-0.05%	12.5 %	173 735
Government and public sector	0.01%	13 %	181 627
Health	-0.00%	2.25 %	30 353
International issues	-0.00%	0.11 %	1 475
Justice, legal system and public safety	-0.15%	13.5 %	187 422
Population and society	-0.03%	7.25 %	102 136
Provisional data	-0.81%	0.25 %	2 350

1. Open government data — case study



Source Track Covid-19 in the U.S., NYT

NYT COVID-19 tracker (2020–2023)

What they did: Compiled case and death counts from 50+ state health departments, thousands of county health agencies, and the CDC — because no single comprehensive source existed. Plus, they published the full dataset on GitHub — enabling hundreds of follow-on stories by other outlets.

Why it mattered: Documented a patchwork of reporting methods. Building on this infrastructure, the NY Attorney General's office found that New York had undercounted nursing home deaths by ~50%.

Lessons learned

- The act of *compiling* data across fragmented sources is itself a journalistic contribution
- Publishing the dataset openly multiplied the impact: hundreds of follow-on stories by other outlets

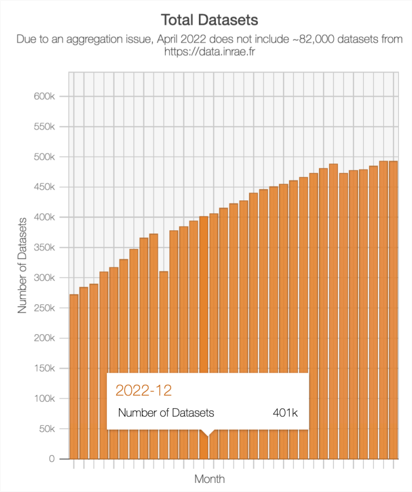
2. Researcher data

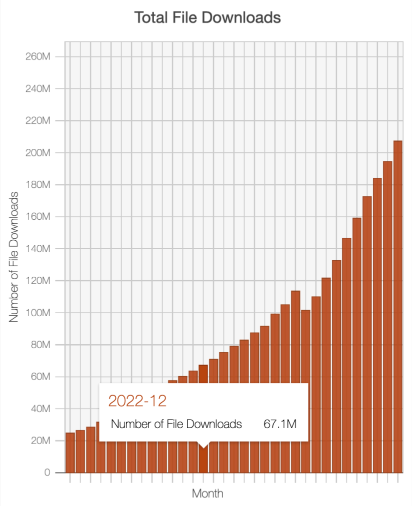
What and why

Datasets compiled and maintained by academics for research purposes — often with high methodological rigor, time-series, cross-national comparability. Extra perk: researchers are usually responsive!

Powerful examples and sources

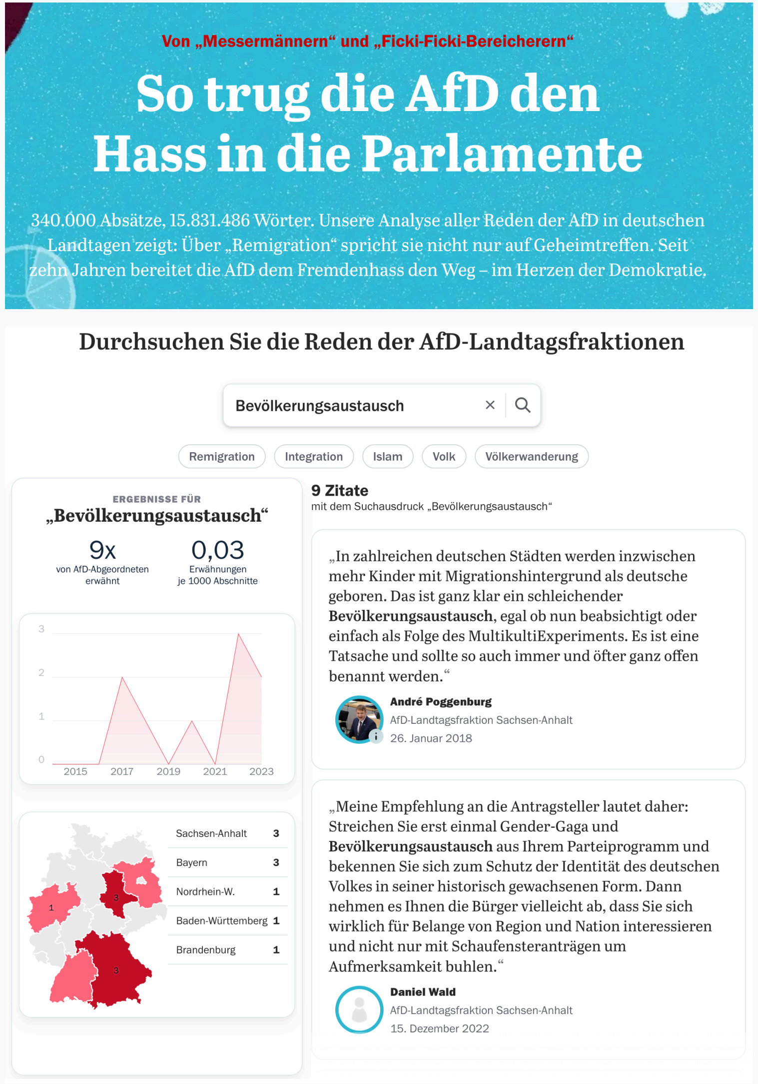
- Harvard Dataverse — replication data for published research
- GDELT Project — global event, media, and network data (massive scale)
- Our World in Data — cleaned, documented data on global development indicators
- Correlates of War — conflict data since 1816
- Manifesto Project — party policy positions across 1000+ parties, 50+ countries
- erikgahner/PolData — curated list of 500+ political science datasets (elections, parties, legislators, conflict, democracy)





`Source` Harvard Dataverse metrics

2. Researcher data — case study



Source Tagesspiegel Interaktiv

Tagesspiegel: AfD hate speech in state parliaments

What they did: Analysed xenophobic speech patterns across all German state parliament debates using StateParl — a research dataset compiled by FU Berlin that converts PDF and Word transcripts from all Landtage into structured, machine-readable text with speaker, state, and faction metadata.

Method: Fine-tuned a language model to classify xenophobic statements; combined with topic modelling to identify dominant narratives and select representative quotes.

Why it's a strong case

- Research infrastructure (StateParl) made an otherwise intractable data collection problem solvable
- Combines computational methods with qualitative validation
- Transparent methodology statement explains both the data source and some analytical choices

3. Survey data

What it is

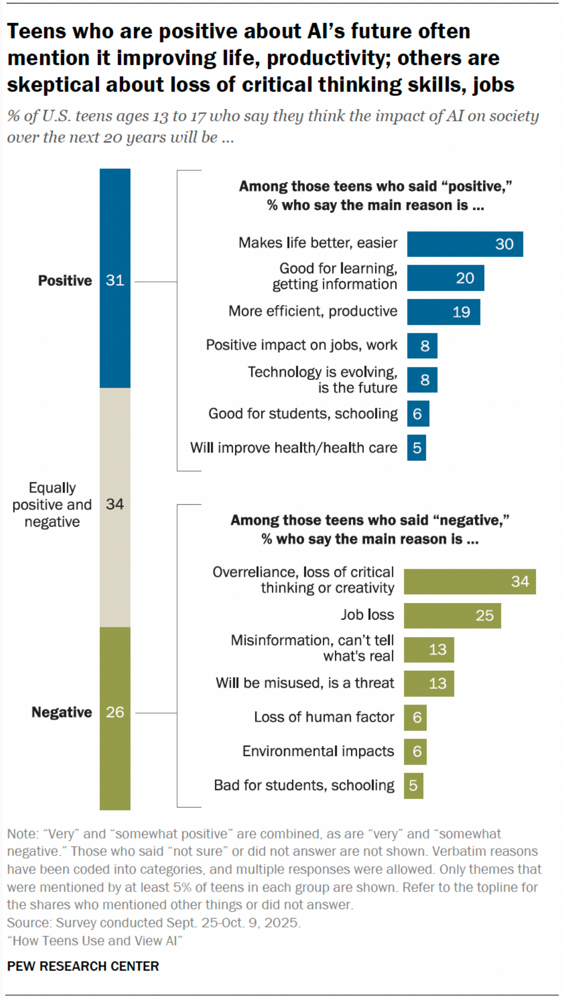
Systematic measurement of attitudes, behaviors, or experiences across a sample of individuals.

Key sources

- Eurobarometer — EU public opinion, twice yearly
- European Social Survey — cross-national, high methods standards
- Pew Research Center — global attitudes, US politics, religion, tech
- ANES — American National Election Studies (since 1948)
- World Values Survey — values and beliefs across 90+ countries
- GESIS — German social science data archive, huge catalogue

Cautions (more next week!)

- Sampling matters: online opt-in panels ≠ probability samples
- Good description is valuable but requires high quality data



`Source` Pew Research Center, 2026

4. Online APIs and algorithmic systems

What it is

Application Programming Interfaces allow programmatic access to data or services. In the age of AI and platform monopolies, APIs are also **windows into automated systems** that make consequential decisions.

Algorithmic/platform APIs worth auditing

- OpenAI moderation endpoint — content classification
- Platform APIs (Reddit, YouTube Data API) — public content and engagement signals
- Ad targeting APIs (Meta, Google) — who gets shown what

The audit opportunity

Algorithmic systems are often opaque. But if they expose an API, you can probe them systematically — varying inputs by race, gender, language, topic — and measure disparate outputs.

`Source` The Markup

5. Web data

What it is

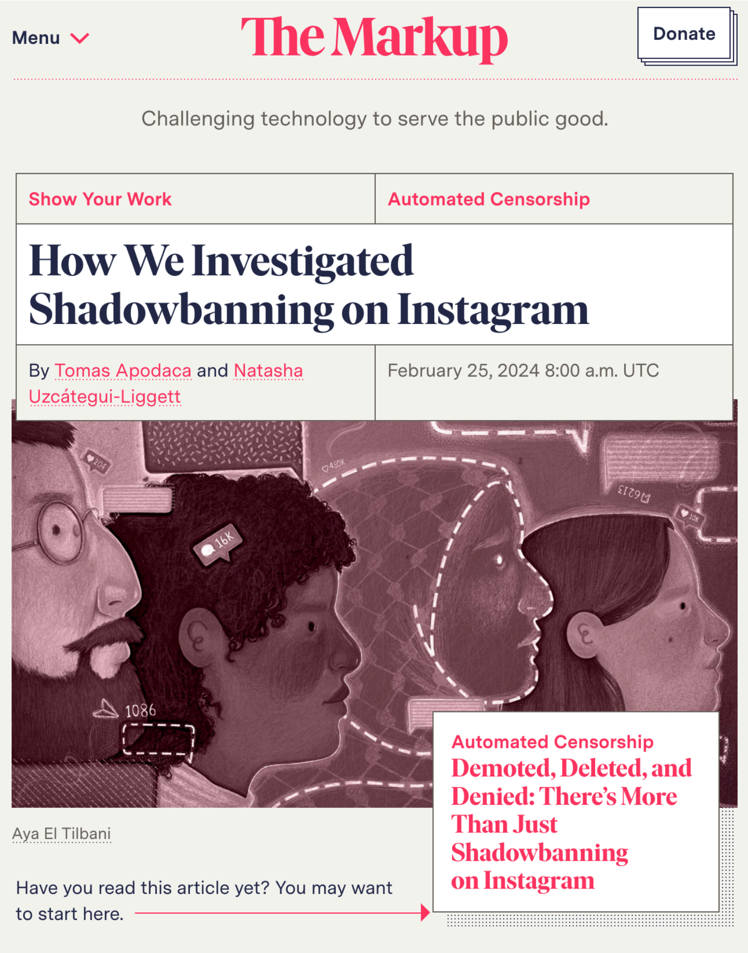
Any data extracted from publicly accessible websites — through scraping (HTML), intercepting undocumented APIs (Network tab), or automated browser interaction (browser automation).

Three modes of access

1. **Static HTML scraping** — `rvest`, `BeautifulSoup`; works when content is server-rendered
2. **Undocumented APIs** — inspect browser Network tab; replicate with `httr2`
3. **Browser automation** — `selenider`, Playwright; for JavaScript-rendered content requiring interaction

What's out there

Company filings, court records, property registries, job postings, social media, price data, entertainment, political advertising, …



`Source` The Markup

6. Investigative and FOIA data

What it is

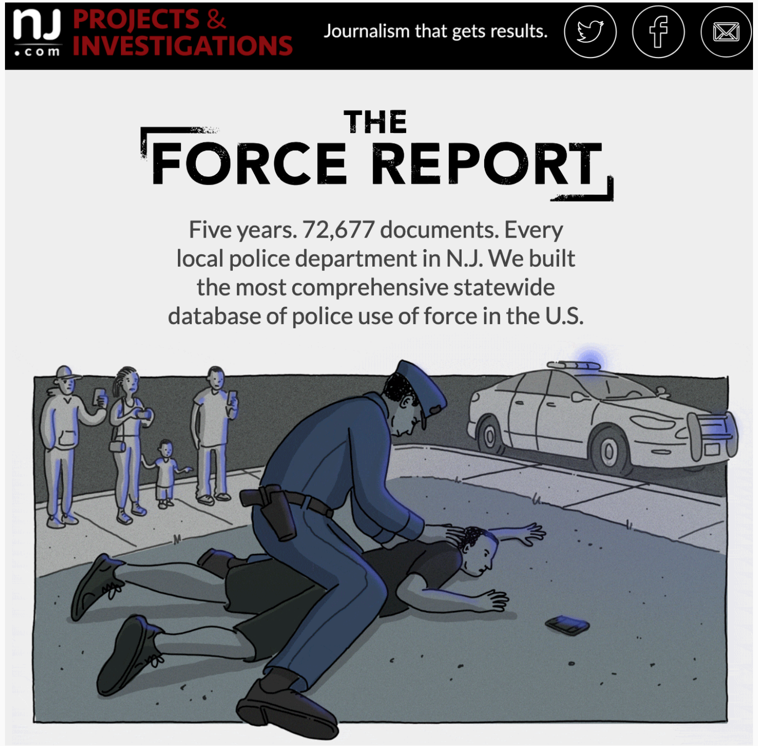
Data obtained through formal legal mechanisms (freedom of information requests), leaks, or through journalistic fieldwork.

FOI mechanisms and resources

- Germany: Informationsfreiheitsgesetz (IFG) — federal;
- EU: Regulation 1049/2001 — access to EU institutions
- US: FOIA — federal; state equivalents vary
- MuckRock — collaborative FOIA filing and publishing platform
- AskTheEU.org — EU equivalent
- FragDenStaat — German equivalent

Why it matters

Some of the most important data in existence is held by governments and not proactively published. FOI turns the *absence* of data into a story — through what is withheld, redacted, or delayed.



`Source` NJ.com, Force Investigation Report

Resources

Where to start

Finding and working with data

- erikgahner/PolData — 500+ political science datasets, well-maintained
- Awesome Public Datasets — broad catalogue across domains
- Google Dataset Search - search engine for datasets across the web
- Our World in Data — cleaned long-run data with source documentation
- Hugging Face Datasets — ML-oriented but contains many useful corpora

APIs

- Public APIs — curated list of free APIs
- RapidAPI Hub — API discovery platform

Investigative data journalism

- GIJN Data Journalism Guide — global investigative journalism network
- The Markup — algorithmic accountability journalism; read for both stories and methodology notes
- Bellingcat Online Investigation Toolkit — OSINT methods

Community and practice

- NICAR-L — data journalism listserv; decades of practical advice
- Datajournalism.com — handbook and case studies
- OpenNews Source — practitioner-written how-tos